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Information security technologies and standards

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What are information security technologies and standards in relation to a database?

The primary aim of data security is to protect the data that an organisation collects, stores, creates, receives or transmits. Compliance is also a major consideration. It doesn't matter which device, technology or process is used to manage, store or collect data – it must be protected.

Data breaches can result in litigation cases, fines and damage to an organisation's reputation. The importance of shielding data from security threats is more important today than it has ever been.

Common threats

Each programming language, framework, and database comes with its own common security threats. It's important to identify these vulnerabilities and develop strategies to mitigate them.

The Open Web Application Security Project (OWASP), an online community, is the definitive source for security threats for web applications. It is regularly updated and clearly defines the top security threats for web applications.

The following table explains common security threats and vulnerabilities that organisations carefully watch out for.

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Threat	Causes
SQL injection	Occurs when malicious code is injected into web-based frontend applications. The code can pass to a backend database and provide threat actors with access to all data stored in the database. We will look at how to mitigate this in a later topic.
Denial of service (DoS)	DoS attacks can slow down the database server or make it unavailable to end users. This will result in poor user experience and significantly high costs in bringing the database back up. This is like many people refreshing your web pages 10,000 times per second at the same time.
Database misconfigurations	Threat actors actively look for misconfigurations as this provides a high chance of exploitation. Databases running with default settings are particularly useful to threat actors, making it easier to hack into these accounts.
Poorly managed sensitive data	Sensitive data is the bread and butter of organisations as well as cybercriminals. It is critical for an organisation to protect their sensitive data. Unfortunately, many organisations do not properly secure their sensitive information, exposing it to attacks.
Weak audit trails	Many regulatory entities require organisations to record and register database events automatically. Keeping track of these events provides an audit trail that enables security teams, database administrators and external auditors to investigate to ensure the data stored within the database is properly protected.

Read ***Database Security Threats and Challenges*** (Mousa et al., 2020)^{fnref:b6d901} to learn about the various security threats and challenges that affect databases and what steps we can take to reduce the risk of these threats to database security.

As you're reading this article, consider the protective methods and countermeasures used to maintain information security in the face of these types of security threats and challenges.

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Once you've read the article Database Security Threats and Challenges, answer the following questions about data security.



☒ Database Security Threats and Challenges

Non-Assessed

In New Zealand, what law protects your personal data and governs how the Open Polytechnic can collect and use your personal data?

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☒ Database Security Threats and Challenges 2

Non-Assessed

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List three types of threats that threaten the security of a database

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☒ Database Security Threats and Challenges 3

[Non-Assessed](#)

Identify the key protection methods that are used to maintain information security

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☒ Database Security Threats and Challenges 4

Non-Assessed

Discuss in detail the countermeasures that are taken to enhance database security.

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The next section will look at essential database security best practices to safeguard sensitive information and ensure system integrity.



Thoughts on this page?

